Kiosk Behavioural Games — Concept Note

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1. Problem statement

We need a short battery of behavioural games, playable at a public kiosk in 10–15 minutes, that elicits observable behaviour linked to 15 target traits and can be validated against established psychometric scales for a population that varies widely in age, literacy, language and familiarity with technology.

Context. This sits within the *Integrated Multimodal Affective-State and Personality Inference Platform* (SHRIII, 24 months). Months 1–3 fixed the construct base on the Big Five and produced a 30-scenario SJT pool. The game strand is still open. Three early concepts exist: MindShift (attention, decision-making, problem solving, flexibility), The Weighmaster (ethics, risk, workplace behaviour) and Table for One (empathy, emotional regulation). Each was designed as a stand-alone desktop game for one developer. None was designed for a kiosk, for speech, or for non-literate players.

Target traits (15). Self-control · emotional regulation · composure under pressure · response consistency · task organisation · decisiveness · adaptability/flexibility · persistence of response style · stability under situational change · expressive restraint vs spontaneity · teamwork · judgement/accountability · problem solving · integrity · initiative and planning/organising.

The core difficulty. The traits are of three different kinds, and one game mechanic cannot carry all of them:

- *Executive and regulatory traits* (self-control, composure, flexibility, problem solving, planning) are best seen in performance under controlled task demands.
- *Style traits* (consistency, persistence of response style, stability under change, decisiveness, expressive restraint) are not single scores at all; they are properties of how a player's behaviour varies across rounds and conditions. They need repeated, comparable trials.
- *Social and moral traits* (teamwork, integrity, judgement/accountability) need another agent, a temptation or a consequence, which means a narrative frame.

Constraints the system must meet.

Constraint	What it means for design
Kiosk, unattended	Self-starting, self-explaining, robust to walk-away; touch plus voice; no keyboard; public space with noise and onlookers
10–15 minutes total	Roughly 8–12 minutes of scorable play after onboarding and consent; every minute must yield data on more than one trait
Preferably 2–3 linked games	Shared world and character so the player feels one experience; game order and handover designed, not left to chance
Wide population (all ages above a floor, all genders, any literacy)	No reading required; cognitive load adjustable; scores interpreted against age and education norms; fairness checked across groups
Multilingual, voice-driven	All instructions and narrative spoken in the chosen language; spoken answers optional, never the only input; icons and gestures carry meaning
Validated against psychometric scales	Game indicators defined in advance and tied to named constructs; a criterion battery administered alongside in pilots; reliability and fairness evidence before any report is released
Report to user, archived	A plain-language, spoken and printed profile; banded rather than precise; consent, retention and deletion rules set before launch

What "working" means. The battery works if, in validation samples, (a) each indicator is reliable enough to interpret (test–retest $r \geq 0.6$ as a first target), (b) indicators relate to their intended scales more than to unrelated ones, (c) scores do not shift by language, literacy or device experience once ability is held constant, and (d) players finish and would play again. Backend storage is out of scope; what the backend receives is in scope, because indicator design depends on it.

Out of scope for this note. Clinical diagnosis, hiring decisions on game scores alone, and any use of camera or voice as a trait signal without separate consent and validation.

2. Review of literature

The literature supports building trait assessment into games, but it also warns that game scores rarely line up neatly with questionnaires. The design must therefore plan for modest convergence, use many short repeated trials, and check fairness across groups from the start.

2.1 What a game-based assessment is

Landers and Sanchez (2022) separate three things that are often confused. A *game-based assessment* infers traits from play itself. A *gamified assessment* is an existing test with game elements added. *Gameful design* uses game thinking to shape any assessment (Landers & Sanchez, 2022). Our brief needs the first, but the most defensible route to it often passes through the second.

The dominant design method is evidence-centred design (ECD), applied to games as *stealth assessment* by Shute and colleagues. ECD links three models: a competency model (what we claim about the person), an evidence model (which behaviours count, and how they are scored) and a task model (situations that elicit those behaviours) (Shute, stealth assessment entry). In *Newton's Playground*, persistence was scored from time on unsolved levels, restarts and revisits, and related to learning gains (Ventura, Shute & Kim, 2013). ECD is the right backbone here because it forces every logged variable to be tied to a named trait before the game is built.

2.2 How well game and task scores agree with questionnaires

This is the central risk. Three bodies of work point the same way.

- **Tasks and surveys measure different things.** In 522 adults given 37 tasks and 23 surveys of self-regulation, task and survey measures showed little relationship. Surveys predicted real-world outcomes with mean R^2 of about 0.10; tasks with about 0.01 (Eisenberg et al., 2019). Dang, King and Inzlicht (2020) argue that self-reports capture typical behaviour across life while tasks capture best performance in one controlled moment (Dang et al., 2020).
- **Robust tasks are often unreliable for individuals.** Classic paradigms such as Stroop, flanker and stop-signal produce strong group effects precisely because people vary little on them, so individual differences have low retest reliability (Hedge, Powell & Sumner, 2018). Reliability, not engagement, is the first thing a kiosk battery is likely to lack.
- **Behavioural risk tasks converge poorly.** Across 1,507 adults, stated risk propensity formed a stable general factor, while behavioural risk tasks such as the BART correlated weakly with each other and with self-report (Frey et al., 2017). Different

BART and Columbia Card Task scores capture different risk constructs (Sambol et al., 2024; Buelow & Blaine, 2015).

Game-based personality studies show the same pattern. Nawaiam, a 20-minute commercial game, correlated with only one Big Five trait (sociability with extraversion, r = .37) in 182 young adults, yet added to the Big Five in predicting adaptive performance (ΔR^2 = .107) (Ramos-Villagrasa & Fernández-del-Río, 2023). Positive cases tend to be narrow: REThink game scores correlated with emotion-regulation measures in youth (David et al., 2024); Coin Catcher scores correlated with morality, altruism and cooperation (Alves et al., 2025); gameplay revealed decision biases that questionnaires missed (Seo, Ryu & Kim, 2018). Machine-learning claims of very high accuracy usually rest on small, self-labelled samples (Elyasi et al., 2026).

Implication. Validation cannot mean "the game reproduces the questionnaire". It should mean the game yields reliable behavioural indicators, these relate to their intended scales more than to unrelated ones, and they add prediction of real outcomes beyond self-report. Reliability must be engineered through many short, comparable trials per indicator.

2.3 Paradigms available for each target trait

Most target traits already have a laboratory paradigm with known properties. The design task is to embed these in play without losing what makes them measure something.

Target trait	Established paradigm	Game-ready indicator	Evidence note
Self-control	Go/No-Go, stop-signal; delay and effort discounting	Commission errors on "don't-touch" items; choosing small-now vs larger-later rewards	Gamified delay discounting kept individual differences (Harvey et al., 2024); effort discounting reliable (Peck & Madden, 2024); inhibition tasks have weak retest reliability (Hedge et al.)
Emotional regulation	Frustration induction; emotion-regulation choice	Response tone after provocation; drift in tone over a run of difficult events	REThink validated against ER scales (David et al., 2024); Table for One's mirroring and drift indices follow this logic

Target trait	Established paradigm	Game-ready indicator	Evidence note
Composure under pressure	Time-stressed decision tasks (Meta-T Tetris)	Change in accuracy and error type when a timer or noise is added	Meta-T isolates time stress as a manipulable variable (Lindstedt & Gray, 2015)
Response consistency	Iowa Gambling Task trial-by-trial learning	Within-person variability of choices on repeated identical situations	Consistency on the IGT shows distinct ERP signatures (Latibeaudiere et al., 2025)
Task organisation	Multi-errand and shopping tasks	Order of sub-tasks, revisits, idle time, resource use	ShopAutiPlan correlated with standard planning tests (Shamhan et al., 2026); stealth facets of conscientiousness (Shute & Ventura, 2013)
Decisiveness	Information-sampling tasks	Time to commit, number of reversals before confirm, information gathered before acting	MindShift logs decision time and "unnecessary changes before confirming"
Adaptability / flexibility	Card-sorting and dimensional-change tasks; reversal learning	Errors and latency after an unannounced rule change	MindShift's rule reversal is a gamified reversal-learning task; Portal 2 gains on flexibility composites (Shute, Ventura & Ke, 2015)
Persistence of response style	Unsolvable or very hard puzzles	Retries, time on hard items, giving up	Persistence from retry logs (Ventura et al., 2013)
Stability under situational change	Same task under toggled conditions	Difference in a player's own indices before vs after a context switch	Meta-T toggles environment variables to test stability (Lindstedt & Gray, 2015)

Target trait	Established paradigm	Game-ready indicator	Evidence note
Expressive restraint vs spontaneity	Nonverbal response inhibition (extra taps)	Extra or premature taps; speed of free vs held responses	Uninhibited extra taps index stopping impulsivity (Ofoe & Anderson, 2021)
Teamwork	Public-goods and cooperation games; collaborative problem solving	Sharing, contribution to a common pot, response to a partner's error	Cooperation in games is stable over about 124 days and matches real-effort cooperation (Peysakhovich, Nowak & Rand, 2014); sharing was the most observable collaborative skill (Wang, 2024)
Judgement / accountability	Sequential moral dilemmas; blame and credit choices	Taking vs sharing credit, owning vs shifting blame, consistency across dilemmas	Players show moral licensing and cleansing across sequential choices (Lee et al., 2023); time pressure shifts moral choices (Holl et al., 2022)
Problem solving	Tower of London; open puzzles	Moves over minimum, planning time before first move, recovery after failure	Tower of London is unidimensional and valid for planning (Debelak et al., 2015); Portal 2 problem-solving gains (Shute et al., 2015)
Integrity	Private-report cheating (die roll, coin flip); bribe offers	Over-reporting when unobserved; accepting a bribe; returning an overpayment	Meta-analysis of 565 studies, 44,050 people: lying varies with situation, gender and age (Gerlach et al., 2019); Coin Catcher (Alves et al., 2025)
Initiative and planning / organising	Open-ended construction; optional side tasks	Optional actions taken without prompt; planning before acting	Minecraft-style open tasks support forward planning (Ameen et al., 2025)

Two things follow from this table. First, the style traits (consistency, persistence of response style, stability under change) are not separate games; they are computed from how the other indicators vary across repeated rounds and a deliberate context change. Second, integrity indicators carry known age and gender effects, so they need norms by group before interpretation (Gerlach et al., 2019).

2.4 Designing for low literacy, many languages and voice

Evidence from India is clear that text fails and that voice with pictures works. With 58 nonliterate and semiliterate adults in Bangalore, a text menu achieved 0% task completion, a spoken-dialogue interface 72%, and a graphical interface with voice-annotated images 100%, though more slowly. The authors recommend combining spoken input with graphical output, so each covers the other's weaknesses (Medhi et al., 2011). Voice-only social platforms such as Sangeet Swara have reached low-income rural users on basic phones (UW, on Vashistha's work).

Speech technology now covers the scale needed. AI4Bharat's IndicConformer supports all 22 scheduled languages, trained on thousands of hours of transcribed Indian speech (AI4Bharat ASR). Recognition of free speech from elderly, accented or code-mixed speakers in a noisy public space will still be imperfect. Speech should therefore be an *option* for answers, with touch on a large icon always available, and spoken answers should be limited to short closed choices.

Translation is not enough for validity. The ITC guidelines require that adapted versions be shown to be equivalent, including checks for differential item functioning across language groups (ITC guidelines, 2nd ed.). For games this extends to narrative: a market inspector, a food stall or a bribe must mean the same thing across regions, ages and genders.

2.5 Gaps this project can fill

- Almost all validated game assessments were built on Western, literate, young samples and screen-and-text interfaces. None we found targets non-literate adults through voice.
- Few studies measure more than three or four traits in one session, and almost none compute style traits (consistency, stability) as within-person variability by design.
- Kiosk administration, which is unsupervised, public and one-shot, is essentially unstudied for personality assessment.

3. Possible pathways

Four pathways are viable; we recommend Pathway A, a three-episode linked game built from embedded classic paradigms, with Pathway B's micro-tasks as its internal mechanics. All four share the same onboarding, voice layer, logging contract and report.

Common foundation (all pathways)

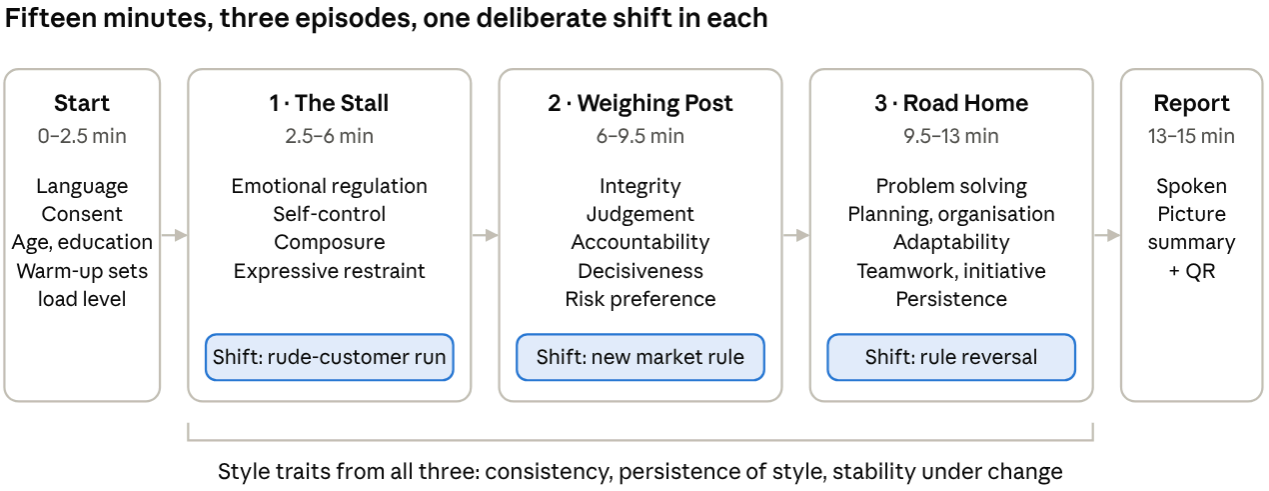
- **Onboarding, 90 seconds.** Language chosen by tapping a face that speaks each language. Consent spoken and confirmed by tap. Age band and education band asked by picture (needed for norms). One practice round that teaches tap, hold and "say or touch".
- **Voice layer.** Every line of narrative and instruction is pre-recorded by native speakers in each language, not synthesised at run-time; this keeps the stimulus identical for everyone in a language. Speech recognition is used only for short closed answers ("yes", "wait", "share") and always mirrors an icon.
- **Calibrated load.** A two-minute warm-up estimates speed and accuracy. The game then sets timer length, number of options and visual clutter in a few fixed levels, not continuously, so scores remain comparable within a level.
- **Repeated structure.** Each indicator is sampled 8–20 times. A deliberate context change (a rule reversal, a timer, a provocation run) is placed in every episode, so stability and consistency can be computed as before–after differences within the person.
- **Logging contract.** Every event carries session, episode, trial, condition, stimulus, response, latency, input mode (voice or touch) and language. Indicators are defined in a scoring dictionary before data collection.

Pathway A — One world, three linked episodes (recommended)

The player spends a day at a village market fair. Three episodes of about 3.5 minutes each flow into one another with the same guide character:

1. **The Stall** (from *Table for One*). Serve a queue of customers whose tone is shown through voice and face. Includes don't-serve items (Go/No-Go), a timed rush (composure) and a run of rude customers (regulation drift). *Traits:* emotional regulation, self-control, composure, expressive restraint.
2. **The Weighing Post** (from *The Weighmaster*). Weigh goods, decide to approve, warn or fine, and handle a bribe, an honest farmer's error and an assistant who makes a mistake or finds fraud. A private-report coin task is embedded as "luck tokens". *Traits:* integrity, judgement/accountability, decisiveness, risk preference.

3. **The Road Home** (from *MindShift*). Plan a route with limited fuel, meet a blocked road, then an unannounced rule reversal. A partner character shares a common fuel pot. *Traits:* problem solving, planning, task organisation, adaptability, teamwork, initiative, persistence.



Pathway A session flow · 15 minutes, 3 episodes

Each episode carries one deliberate shift, so stability and consistency are read as the change in a player's own behaviour before and after it.

Style traits are computed across all three episodes. *Strength:* one story, strong engagement, reuses existing concepts. *Risk:* the most build effort, and episodes are harder to reorder.

Pathway B — Micro-game carousel

Six to eight 60–120-second mini-games, each a lightly dressed classic paradigm (balloon pump, don't-tap, card sort with switch, tower puzzle, share-the-pot, private dice), joined by a simple frame. A selector picks the next mini-game to reduce uncertainty on the least-measured traits. *Strength:* closest to validated paradigms, easiest to validate trait by trait, easy to swap modules. *Risk:* feels like a test, weaker on integrity and accountability, which need context and consequence.

Pathway C — Voiced scenario game (SJT-game hybrid)

The 30 Big Five situational scenarios become short illustrated, voiced scenes with timed choices and consequences that carry over. Two micro-tasks are inserted for executive traits. *Strength:* builds directly on the Month 1–3 SJT work, maps cleanly to the Big Five, low build cost. *Risk:* closer to a self-report in disguise and more open to faking; weakest for style and executive traits.

Pathway D — Co-play kiosk (add-on)

Two people play side by side at a wide screen, or one player with a scripted partner. Measures teamwork, turn-taking, credit and blame directly. *Strength:* the only pathway that observes real social behaviour. *Risk:* pairing is not controllable in a public space; best kept as an optional fourth episode or a supervised-site variant.

Trait coverage by pathway

● strong indicator · ◐ partial · ○ weak or none

Trait	A: Linked episodes	B: Carousel	C: Voiced scenarios	D: Co-play
Self-control	●	●	◐	○
Emotional regulation	●	◐	◐	◐
Composure under pressure	●	●	◐	◐
Response consistency	●	●	◐	○
Task organisation	●	◐	○	◐
Decisiveness	●	●	◐	◐
Adaptability / flexibility	●	●	◐	◐
Persistence of response style	●	●	○	○
Stability under situational change	●	◐	○	○
Expressive restraint vs spontaneity	◐	●	○	◐
Teamwork	◐	◐	◐	●
Judgement / accountability	●	○	●	●
Problem solving	●	●	◐	◐
Integrity	●	◐	◐	◐
Initiative and planning / organising	●	◐	◐	◐
Build effort	High	Low	Low	Medium
Ease of validation	Medium	High	High	Low

Suggested sequence. Build Pathway B's mini-games first as the validated core, then wrap them into Pathway A's three episodes. This gives a fallback if the narrative version loses reliability, and lets the team compare the two directly in the pilot.

4. Validation, reporting and open questions

No report should reach a player until the battery has passed reliability and fairness checks in at least two language groups.

Validation stages

Stage	Sample (indicative)	What it establishes
1. Think-aloud and observation	30–40, spread across age, literacy and 3 languages	Instructions understood; narrative means the same across groups; no item depends on reading
2. Pilot with criterion battery	300–400 per language, quota by age, gender, education	Indicator distributions; internal consistency; convergent and discriminant correlations with scales
3. Retest	100 of the pilot, 2–4 weeks later	Retest reliability per indicator; which indicators to keep
4. Fairness and equivalence	Pooled pilot data	Measurement invariance and differential item functioning across language, literacy, age band and gender (ITC guidelines)
5. Criterion and norming	Larger field sample, with an outcome (supervisor or peer rating, or an observed task)	Incremental prediction beyond self-report; norms by age and education band

Criterion measures to pair with the games

These are standard, short, widely translated scales; Indian-language versions must be checked or adapted. For players who cannot read, all scales are read aloud by a trained administrator in the pilot only, never at the public kiosk.

Game construct	Candidate criterion scale
Big Five base (all traits)	BFI-2-S or BFI-2-XS
Self-control	Brief Self-Control Scale
Emotional regulation, composure	Emotion Regulation Questionnaire; DERS-16
Adaptability / flexibility	Cognitive Flexibility Inventory; card-sorting task score
Persistence	Short Grit Scale
Integrity, judgement	HEXACO Honesty–Humility; moral-identity items

Game construct	Candidate criterion scale
Decisiveness	General Decision-Making Style scale
Teamwork	Peer or partner rating after co-play
Problem solving, planning	Tower of London score
Response style traits	No direct scale; validated through retest and theoretical patterns

Because self-report and behaviour converge weakly (Section 2.2), the target is a correlation pattern, not a level: each game indicator should correlate more with its own scale than with others, typically in the r = .2–.4 range.

The player's report

- Spoken in the player's language at the kiosk, with a one-page picture summary printed or sent by QR code.
- Banded and strengths-framed (for example "steady under pressure", "prefers to plan before acting"), never a single number or a label such as "dishonest".
- States that it is a behavioural snapshot from one session, not a diagnosis.
- Integrity and risk results are reported to the player only as broad styles, or held back entirely, until their validity is established.

Data and consent

Consent is spoken and confirmed by tap, with a separate opt-in for archiving and for any later contact. Data handling must follow India's Digital Personal Data Protection Act, 2023, including purpose limitation and the right to erasure. Audio recordings of speech are not kept by default; only the recognised answer is logged.

Open questions

- [ ] What is the minimum age? Card-sorting and inhibition paradigms work from early school age, but integrity and accountability scenarios need a higher floor.
- [ ] Which three languages for the first pilot?
- [ ] Will kiosks sit in supervised settings (colleges, employment offices) or open public spaces? This decides whether Pathway D is feasible.
- [ ] What decisions will the report inform? Any use in selection raises the evidence bar and the legal exposure considerably.
- [ ] Is camera or voice-tone analysis (from the wider multimodal platform) in scope for the games, or kept separate?

Main risks

- **Low reliability** of short behavioural indicators (Hedge et al., 2018). *Mitigation:* many trials per indicator; drop indicators that fail retest.
- **Cultural mismatch** of scenarios such as bribes, markets or gender roles. *Mitigation:* think-aloud in each region; ITC adaptation.
- **Speech errors** for elderly, accented or code-mixed speakers in noise. *Mitigation:* touch always available; voice used only for closed choices.
- **Gaming the game** once players learn what is scored. *Mitigation:* no visible score; randomised variants; integrity tasks kept private-report.

References

Project materials: Three-Month Progress Report (31 Aug 2026); MindShift Initial Game Plan (R. Robert); The Weighmaster and Table for One design reports; Common Literature Survey (39 entries). Papers from the team's survey are cited in the text by DOI.

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